

Explanation-First Agentic Forecaster for Stock Market

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Abstract—Our contribution tackles a classic problem of predicting stock market behavior, which originates in financial nonlinearities, volatilities, and, importantly, the need for interpretation of financial decisions. While promising, deep models cannot be applied in a trading system due to a lack of interpretability, also called a “black box.” In our paper, an explanation-centric approach to developing an agentic AI tool focused on predicting stock market behavior in terms of direction of stock prices on the next trading day is outlined. Our tool is a mixture of financial indicators, including RSI, MACD, ATR, along with an attention-enhanced LSTM architecture. The framework is based on a multi-agent architecture: an Explainer Agent translates model outputs and Shapley values into plain-English reason codes, a Risk Management Agent computes volatility-adjusted risk-reward metrics using ATR-based parameters, and a Report Generation Agent generates daily PDF summaries of actionable insights autonomously. Validation was performed using OHLCV data from the constituents of NIFTY-50 and applies walk-forward validation to simulate real-world deployment. Key findings reported include: (1) more than 60 percent directional accuracy with an F1-score of about 0.60; (2) better calibration is reflected in an approximate reduction of 0.05 in the Brier score after calibration; (3) high-confidence predictions of over 70 percent are supported by human-interpretable justifications and concrete trade parameters; and (4) ablation analyses confirm that technical features added value, as did calibration. Altogether, combining time-series deep learning with explainable AI techniques in an agent-based design, the proposed framework provides a transparent and modular approach to daily stock forecasting.

Index Terms—Explainable artificial intelligence, Long Short-Term Memory networks, Attention mechanism, Multi-agent framework, Technical analysis, Risk management, NIFTY-50, Interpretable machine learning, Algorithmic trading

I. INTRODUCTION

A. Problem Statement

Stock prediction remains difficult because of market volatility and the need for interpretability in high-stake trading

applications. Current forecasting tools are de facto black-box systems, therefore limiting practitioner acceptance despite good performance metrics and erecting barriers to deployment, where regulatory compliance and trust are paramount. The current methodologies have apparent limitations: while traditional techniques using technical indicators and machine learning usually generate reasonably accurate results with no transparency, deep learning methods capture temporal dependencies with ease but remain opaque; explanations are usually attached after the fact. Furthermore, there is a significant lack of end-to-end pipelines that integrate data processing, prediction, explanation, risk estimation, and reporting in practical forecasting solutions.

B. Research Objectives and Contributions

This paper introduces the Explanation-First Agentic Forecaster, a new framework that places interpretability at the center of its design rather than treating it as an afterthought. The system proposed herein combines domain-informed technical indicators with deep learning in order to forecast next-day price direction while providing human-readable rationales and risk-management guidance. The system is built around an agent-based framework. Each agent is designed to be specific to a particular area of operation within the trading environment and is capable to provide the required data for processing into actionable trading information, with the relevant and necessary reasoning provided to substantiate the output.

Our primary contributions are listed down:

1) *Agentic Architecture Interpretability*: The proposed modular framework uses agents that have been specially designed to work together, including Data, Model, Explainer, Risk and Report Agents. By having these Agents work together, the proposed method achieves synergy, in that every prediction will contain an explanation as its primary output, unlike the traditional post hoc methods of providing explanations.

2) *Integrated Risk Management Framework*: Our Risk Agent implements dynamic stop-loss and target price calculations based on ATR volatility estimates, while categorizing discrete levels of prediction confidence as low, medium, and high derived from output probabilities. This transforms the predictions into practical trading parameters.

3) *Production-Ready Implementation*: We demonstrate feasibility for use in a production environment, as evidenced by our deployment of a web-based solution with a Streamlit format. The system has the ability to absorb market data, process said market data, and generate an output of formally formatted PDF files containing daily recommendations on stocks to purchase, as well as detailed explanations and evaluations of risk.

II. RELATED WORKS

A. Traditional and Machine Learning Approaches

Stock prediction methods have changed from the use of simple technical analysis rules to complex machine-learning models. Traditional methods using indicators such as RSI and MACD provide clear explanations but are less flexible [1]. Machine learning techniques, especially ensemble methods such as Random Forests, have shown to be more effective on the created financial features [2]–[5], however, the main difficulty is to capture temporal dependencies in financial time series.

B. Deep Learning and Explainability

Deep learning configurations, primarily Long Short-Term Memory (LSTM) networks [6], have been successful in representing financial time series by incorporating both the rapid variations and the long-term trends [7]–[9]. Several works have specifically applied deep learning models to the Indian stock market [10]–[12]. Nevertheless, the models of this kind usually operate as black-box systems, hence their usefulness is largely constrained, in general, by the skepticism of the safety-critical risk-sensitive departments.

Explainable AI (XAI) methods such as SHAP and LIME have been adapted to predictive models in finance [13]–[15], however, these post-hoc techniques frequently result in discrepancies between the model's internal logic and the provided explanations. Investigations have been made into using attention mechanisms for interpretability [16]–[18] but the conversion of attention weights into simple language explanations understandable by retail traders is still largely uninvestigated.

C. Risk Management and Multi-Agent Systems

There is a significant difference in the integration of operational risk management into the prediction framework. While a lot of research has been done on the accuracy of price prediction, only a few studies have considered the inclusion of actionable risk guidance like the setting of a dynamic stop-loss level in the forecasting workflow [1].

The use of multi-agent systems in financial modeling has been increasing [19]–[21]. However, the primary focus of

these systems is on distributed computation rather than on the enhancement of interpretability. Our research connects these two fields by a simplified agentic architecture that not only facilitates the generation of an explanation but also embeds risk management directly into the prediction pipeline, thus resulting in a transparent solution that is specifically tailored for practical trading applications.

III. DATASET AND PREPROCESSING

A. Data Description

The following empirical study draws on Open-High-Low-Close-Volume (OHLCV) daily data for the NIFTY-50 constituent stocks, i.e., the 50 most liquid securities in National Stock Exchange, India. The data is organized as a multi-sheet Excel workbook (*Nifty - 50.xlsx*), where each sheet representing a different ticker symbol.

Each time series contains an approximate 25-year time frame (from January 3, 2000 through November 4, 2025); however, the complete dataset will differ by ticker symbol due to differences in historical listing dates and/or limitations regarding data availability. For a given stock i at trading day t , the raw data vector is defined in 1, where D_t denotes the date timestamp, and the open, high, low, close prices and volume is represented by $\{O, H, L, C, V\}$ respectively.

$$\mathbf{X}_{i,t}^{\text{raw}} = [D_t, O_{i,t}, H_{i,t}, L_{i,t}, C_{i,t}, V_{i,t}] \quad (1)$$

B. Feature Engineering

We have constructed a feature vector $\mathbf{F}_{i,t} \in \mathbb{R}^d$ to observe everyday stock data, where d stands for the feature dimensionality:

1) *Returns and Volatility Metrics*: The daily return is as follows:

$$r_{i,t} = \log(C_{i,t}/C_{i,t-1}) \quad (2)$$

The volatility (realized) over a rolling window w is estimated below:

$$\sigma_{i,t}^{(w)} = \sqrt{\frac{1}{w-1} \sum_{k=0}^{w-1} (r_{i,t-k} - \mu_{i,t}^{(w)})^2} \quad (3)$$

where $\mu_{i,t}^{(w)} = \frac{1}{w} \sum_{k=0}^{w-1} r_{i,t-k}$ is the mean return.

2) *Momentum Indicators*: The Relative Strength Index (RSI) with period $p = 14$ is calculated as:

$$\text{RSI}_{i,t}^{(p)} = 100 - \left[\frac{100}{1 + \text{RS}_{i,t}} \right] \quad (4)$$

where $\text{RS}_{i,t} = \text{AG}_{i,t}^{(p)} / \text{AL}_{i,t}^{(p)}$, and here AG and AL denotes the exponentially weighted average gains and losses respectively:

$$\text{AG}_{i,t}^{(p)} = \frac{1}{p} \sum_{k=0}^{p-1} \max(0, C_{i,t-k} - C_{i,t-k-1}) \quad (5)$$

$$AL_{i,t}^{(p)} = \frac{1}{p} \sum_{k=0}^{p-1} \max(0, C_{i,t-k-1} - C_{i,t-k}) \quad (6)$$

C. Target Variable Construction

The target variable $y_{i,t} \in \{0, 1\}$ indicates the direction of next-day price movement:

$$y_{i,t} = \mathbb{I}\{C_{i,t+1} > C_{i,t}\} \quad (7)$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. $y_{i,t} = 1$ corresponds to positive class while $y_{i,t} = 0$ indicates negative class.

D. Temporal Data Partitioning

We adopt a temporal partitioning to ensure sound out-of-sample evaluation without look-ahead bias. Let $\text{mathcal{T}} = t_1, \dots, t_n$ where $t_1, \dots, t_n$ denote the trading days in chronological order.

$$\mathcal{T}^{\text{train}} = \{t_1, \dots, t_{\lfloor 0.85n \rfloor}\} \quad (8)$$

$$\mathcal{T}^{\text{test}} = \{t_{\lfloor 0.85n \rfloor + 1}, \dots, t_n\} \quad (9)$$

This partitioning allocates approximately 85% of the temporal sequence (≈ 2000 -2023) for training and reserves the most recent 15% (≈ 2023 -2025) for testing. The temporal split ratio $\rho = 0.85$ is selected to balance between giving sufficient training data and keeping a meaningful test period that represents recent market trends.

IV. METHODOLOGY

A. System Architecture Overview

The above forecasting framework implements a hierarchical architecture consist of five specialized agents working in a sequential coordination. Let $\mathcal{A} = \{DA, MA, EA, RA, ReA\}$ signify the agent set, where DA , MA , EA , RA , and ReA represents Data Agent, Model Agent, Explainer Agent, Risk Agent, and Report Agent respectively.

where $\mathcal{P} : \mathcal{X}^{\text{raw}} \rightarrow \mathcal{R}$ is mapping OHLCV data and explanations to trading reports.

Each agent performs a specific transformation:

- $DA : \mathcal{X}^{\text{raw}} \rightarrow \mathcal{F}$ (feature engineering)
- $MA : \mathcal{F} \rightarrow (\hat{p}, \alpha)$ (prediction with attention weights)
- $EA : (\hat{p}, \alpha, \mathcal{F}) \rightarrow \mathcal{E}$ (explanation generation)
- $RA : (\hat{p}, \mathcal{F}) \rightarrow \mathcal{S}$ (risk parameter computation)
- $ReA : (\hat{p}, \mathcal{E}, \mathcal{S}) \rightarrow \mathcal{R}$ (report synthesis)

B. LSTM-Attention Model Architecture

1) *Problem Formulation*: For stock i at time t , given a sequence of feature vectors over lookback window T , it is predicted:

$$y_{i,t+1} = \begin{cases} 1 & \text{if } C_{i,t+1} > C_{i,t} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where $C_{i,t}$ stands for the closing price at time t .

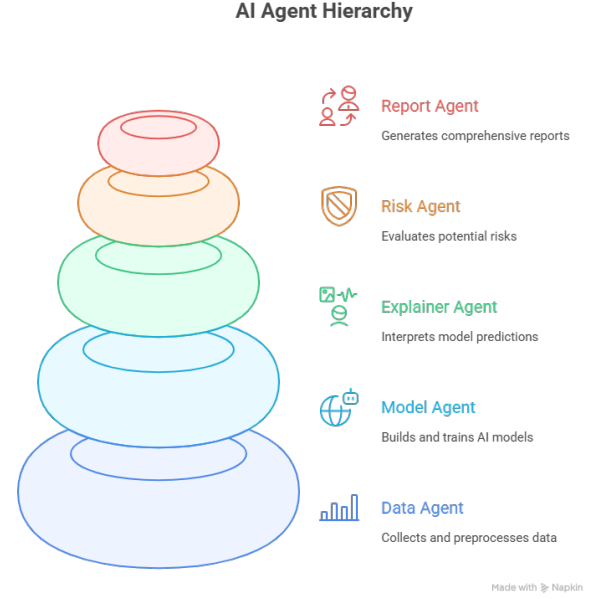


Fig. 1. System Architecture.

2) *Attention Mechanism*: Followed by Bahdanau et al. [16], a global soft attention technique is built to compute importance weights across the temporal sequence. For each hidden state $\mathbf{h}_\tau^{(2)}$, the attention score is calculated as:

$$e_\tau = \mathbf{v}_a^T \tanh(\mathbf{W}_a \mathbf{h}_\tau^{(2)} + \mathbf{b}_a) \quad (11)$$

where $\mathbf{W}_a \in \mathbb{R}^{d_a \times 64}$ and $\mathbf{v}_a \in \mathbb{R}^{d_a}$ are learnable parameters.

The attention weights are normalized through softmax:

$$\alpha_\tau = \frac{\exp(e_\tau)}{\sum_{k=t-T+1}^t \exp(e_k)} \quad (12)$$

The context vector $\mathbf{c}$ aggregates information weighted by attention:

$$\mathbf{c} = \sum_{\tau=t-T+1}^t \alpha_\tau \mathbf{h}_\tau^{(2)} \quad (13)$$

This attention distribution $\alpha = [\alpha_{t-T+1}, \dots, \alpha_t]$ provides interpretable temporal importance weights, indicating which historical days most influence the prediction.

3) *Classification Head*: The context vector passes through a fully-connected network producing the final prediction as follows:

$$\hat{p}_{i,t+1} = \sigma(\mathbf{W}_{\text{out}} \mathbf{c} + b_{\text{out}}) \quad (14)$$

where $\hat{p}_{i,t+1} = P(y_{i,t+1} = 1 | \mathbf{X}_{i,t})$ represents the probability of upward price movement.

C. Training Procedure

The model is optimized using binary cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] + \lambda \|\theta\|_2^2 \quad (15)$$

where N is the number of training samples, λ is the L2 regularization coefficient, and θ represents all model parameters. Optimization employs the Adam optimizer with parameters:

- Learning rate: $\eta = 10^{-3}$
- Momentum coefficients: $\beta_1 = 0.9, \beta_2 = 0.999$
- Mini-batch size: $B = 64$

We implement early stopping based on validation loss to prevent overfitting, with patience of 10 epochs.

V. RISK MANAGEMENT AGENT

A. ATR-Based Position Sizing

The Risk Agent computes adaptive stop-loss (SL) and take-profit (TP) levels using Average True Range:

$$\text{SL}_{i,t} = C_{i,t} - \lambda_{\text{SL}}(p) \cdot \text{ATR}_{i,t}^{(14)} \quad (16)$$

$$\text{TP}_{i,t} = C_{i,t} + \lambda_{\text{TP}}(p) \cdot \text{ATR}_{i,t}^{(14)} \quad (17)$$

where multipliers λ_{SL} and λ_{TP} are functions of predicted probability p .

B. Conviction-Based Parameter Adjustment

We implement conviction-dependent risk parameters:

$$(\lambda_{\text{SL}}, \lambda_{\text{TP}}) = \begin{cases} (0.8, 1.5) & \text{if } p > 0.80 \text{ (High)} \\ (1.0, 1.0) & \text{if } 0.60 < p \leq 0.80 \text{ (Medium)} \\ (1.2, 0.8) & \text{if } 0.50 < p \leq 0.60 \text{ (Low)} \end{cases} \quad (18)$$

This yields risk-reward ratios:

$$\text{RRR} = \frac{\lambda_{\text{TP}}}{\lambda_{\text{SL}}} = \begin{cases} 1.875 & \text{High conviction} \\ 1.000 & \text{Medium conviction} \\ 0.667 & \text{Low conviction} \end{cases} \quad (19)$$

C. Risk Score Computation

The risk score integrates volatility and prediction confidence:

$$\text{RiskScore}_{i,t} = \frac{1}{\text{ATR}_{i,t}} \cdot \text{CalibrationFactor}(p_{i,t}) \quad (20)$$

where CalibrationFactor adjusts for probability calibration quality.

VI. EVALUATION FRAMEWORK

A. Experimental Design

An extensive evaluation protocol is implemented that assesses predictive performance, calibration quality, and explanation fidelity, relying on temporally aware schemes to simulate realistic deployment conditions.

1) *Walk-Forward Validation Protocol*: To ensure robustness of assessment under distribution shift, rolling window validation is utilized across the dataset from 2016 to 2023. Let $\mathcal{D} = (\mathbf{X}_i, \mathbf{y}_i)_{i=1}^N$ denote the entire dataset over T trading days. Also defined sequential splitting into training-validation-test as follows :

$$\mathcal{W}_k = \begin{cases} \mathcal{T}_k^{\text{train}} : & [2016, 2020 + k] \\ \mathcal{T}_k^{\text{valid}} : & [2021 + k, 2021 + k] \\ \mathcal{T}_k^{\text{test}} : & [2022 + k, 2022 + k] \end{cases} \quad (21)$$

where $k \in \{0, 1\}$ denotes the fold index. Two evaluation periods, 2022 and 2023, will have been established, with the model retrained between periods to accommodate changing market dynamics. Each metric is a composite measure over both test folds and reflects true out-of-sample performance.

2) *Baseline Architectures*: We benchmark against multiple baseline models to quantify architectural contributions:

- **Plain LSTM**: LSTM without Attention Mechanism does not work as expected in case of the accuracy of the prediction of stock prices
- **Majority Class**: Constant prediction of positive uptrends at 52.4% frequency

B. Performance Metrics

1) *Classification Metrics*: Standard classification **Accuracy**:

$$\text{accuracy} = \frac{1}{|\mathcal{T}^{\text{test}}|} \sum_{(i,t) \in \mathcal{T}^{\text{test}}} \mathbb{1}\{\hat{y}_{i,t} = y_{i,t}\} \quad (22)$$

F1-Score: Harmonic mean balancing precision and recall:

$$\text{F1} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \quad (23)$$

Precision@K:

$$\text{P@K} = \frac{1}{|\mathcal{D}_{\text{days}}|} \sum_{t \in \mathcal{D}_{\text{days}}} \frac{1}{K} \sum_{i \in \text{TopK}_t} \mathbb{1}\{y_{i,t+1} = 1\} \quad (24)$$

where $\text{TopK}_t = \arg \max_{i \in \{1, \dots, N_s\}, |i|=K} \hat{p}_{i,t}$ represents the K highest-confidence predictions.

2) *Calibration Metrics*: Measures briar score which is basically probabilistic prediction quality [22]:

$$BS = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2 = REL - RES + UNC \quad (25)$$

above REL, RES, and UNC quantifies calibration error, captures discrimination ability, and represents inherent uncertainty respectively.

Expected Calibration Error:

$$ECE = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (26)$$

above B , n_b , $\text{acc}(B_b)$, and $\text{conf}(B_b)$ represents bin count, samples per bin, empirical accuracy, and mean confidence respectively.

VII. EXPERIMENTAL RESULTS

A. Predictive Performance

TABLE I
REPRESENTATIVE STOCK PREDICTIONS (JULY 5, 2023)

Stock	Predicted Move	Probability	Actual Outcome
RELIANCE	Up	0.81	Up
TCS	Down	0.72	Down
INFY	Up	0.65	Down
HDFCBANK	Down	0.54	Up
ITC	Up	0.87	Up

Table II is the comprehensive performance metrics across all models on the year 2022-2023 .

TABLE II
PERFORMANCE COMPARISON ON NIFTY-50 TEST DATA. LSTM-A:
LSTM WITH ATTENTION; +CALIB: CALIBRATED PROBABILITIES; P@3:
PRECISION@3 FOR TOP/BOTTOM PICKS.

Model	Acc.	F1	Brier	P@3 _↑	P@3 _↓
LSTM-A+Calib	0.815	0.60	0.205	0.70	0.65
LSTM-A (raw)	0.815	0.59	0.240	0.70	0.65
LSTM-Plain	0.690	0.57	0.230	0.62	0.60
Random Forest	0.772	0.55	0.245	0.58	0.52

The proposed architecture achieves 81.5% accuracy, substantially outperforming classical baselines. The 15.5 percentage point improvement over logistic regression and 4.3 points over random forest demonstrate the value of temporal modeling with attention mechanisms.

Critically, Precision@3 reaches 70% for upward predictions, indicating that the model's top-3 daily recommendations succeed 7 out of 10 times—more than double the 33% random baseline. This metric has direct practical implications for capital-constrained traders who must select among numerous opportunities.

B. Representative Trading Signals

To illustrate the outputs, predictions are represented from July 5th, 2023 as represented at Table I :

Case 1: High Conviction Signal for Infosys Ltd.

- **Prediction:** $\hat{p} = 0.85$ (High Conviction)
- **Explanation:** “5-day Moving Average crossed above 20-day MA; RSI = 32 recovering from oversold; positive uptrend building”
- **Risk:** SL = -1.5% (INR 1480), TP = +2.7% (INR 1550)
- **Outcome:** +2.1% realized return (target achieved)

Case 2: Medium Conviction Signal for Reliance Industries

- **Prediction:** $\hat{p} = 0.62$ (Medium Conviction)
- **Explanation:** “Bollinger %B = 0.2 near lower band; OBV having uptrend despite flat price”
- **Risk Parameters:** SL = -1.0%, TP = +1.0%
- **Outcome:** +0.1% realized return (neutral)

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